

Retail Sales Analytics Dashboard

End-to-End Data Analytics Project using Python, SQL, and Power BI

Business Objective

- Analyze retail sales performance across regions and time periods
- Identify sales trends, seasonality, and growth patterns
- Compare regional performance to optimize resource allocation
- Understand customer purchasing behavior and preferences
- Improve profitability using data-driven insights and recommendations



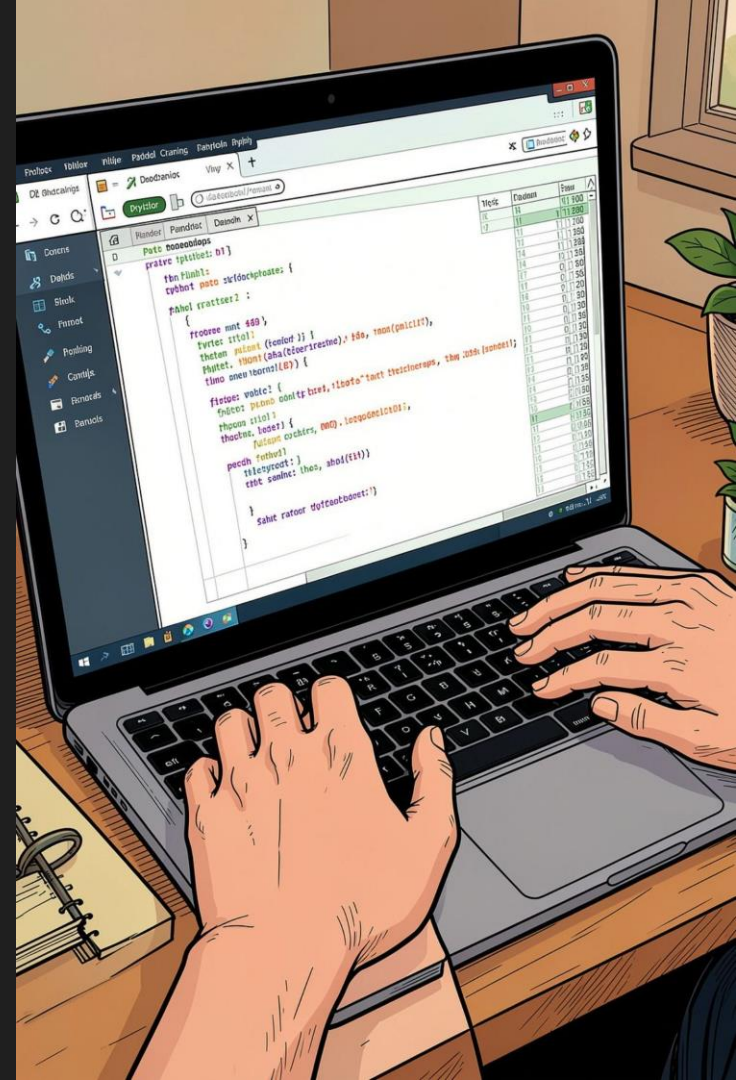


Dataset Overview

- Source: Kaggle
- Records: 51,290
- Columns: 26
- Orders, Customers, Products
- Sales, Profit, Order Date
- Region, Segment, Category

Data Preparation

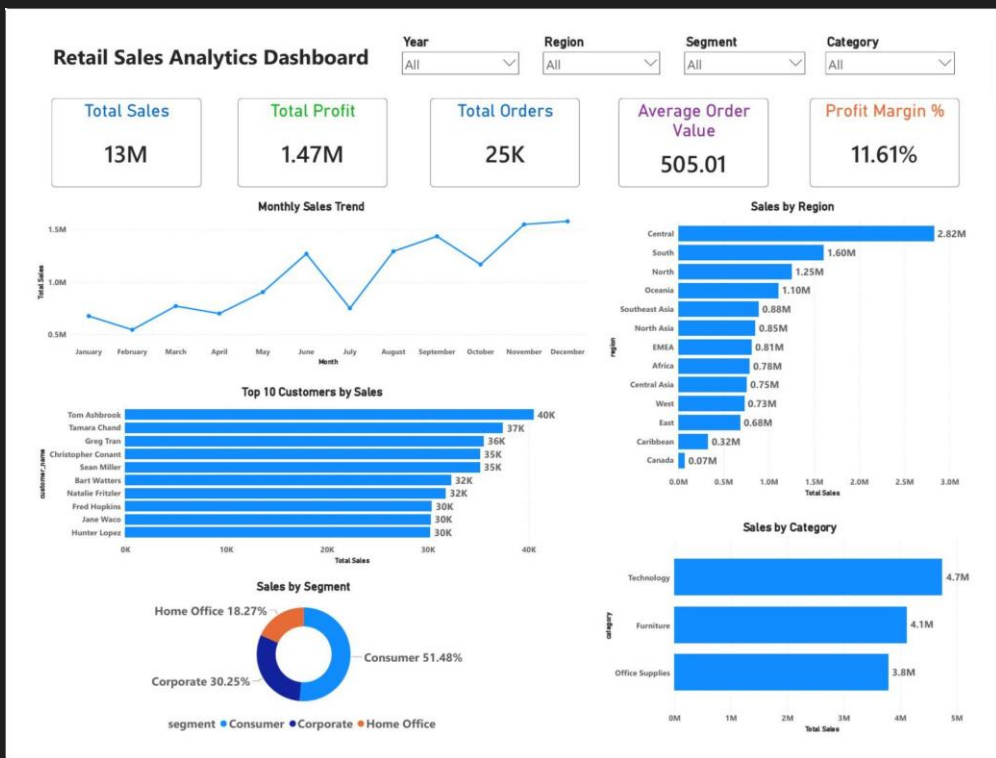
- Data Quality Assessment
- Missing Value Handling
- Duplicate Removal
- Data Type Conversion
- Data Cleaning using Python (Pandas)





SQL Analysis

- **Regional Sales Analysis**
- **Monthly Sales Trends**
- **Top Customers by Revenue**
- **Category Performance**
- **KPI Calculations**



Power BI Dashboard

- Interactive Dashboard
- KPI Cards
- Sales Trend Analysis
- Regional Performance
- Customer & Category Analysis

Key Business Insights

- **Highest-performing Region:** Central Region generated the highest sales (2.82M).
- **Best-selling Category:** Technology was the highest-selling category (4.7M).
- **Most Profitable Segment:** Consumer Segment contributed the highest share of profit (51.06%).
- **Peak Sales Month(s):** Sales peaked in December, with another strong peak in November.
- **Business Recommendation:** Increase investment in high-performing regions and Technology products while improving performance in lower-performing regions.

Conclusion & Next Steps

- Completed an end-to-end Retail Sales Analytics project
- Performed data cleaning using Python
- Solved business problems using SQL
- Built an interactive Power BI dashboard
- Improved analytical and visualization skills

Thank You

Questions?

